

Principles of Economics

Problem Set 4

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. Think of a job with which you are familiar, perhaps a teacher, a retail worker, a nurse, or a police officer. Indicate why the employment contract is necessarily incomplete. What important parts of the person's job—things that the employer would like to see the employee do or not do—cannot be covered in a contract, or if they are, cannot be enforced?
2. For the following example, explain who the principal is, who is the agent, and what aspects of their interaction are of interest to each and are not covered by a complete contract. A charity wants to commission research to find out as much as possible about a new virus.
3. Read the following statements about Figure 6.6 and choose the correct option(s).
 - a) The vertical axis intercept of the hiring line (top panel), €550, represents the reservation wage of all the workers.
 - b) ii. The hiring line (top panel) shows the wage required to hire the number of workers needed to maintain the firm's workforce.
 - c) iii. The point (70, €725) on the reservation wage curve (bottom panel) means that when 70 workers are employed, their reservation wages are at least €725.
 - d) iv. If the quitting rate becomes 10% (instead of 4%) and the firm still needs to maintain a workforce of 50, then *ceteris paribus*, the wage it needs to set will be higher than €675.

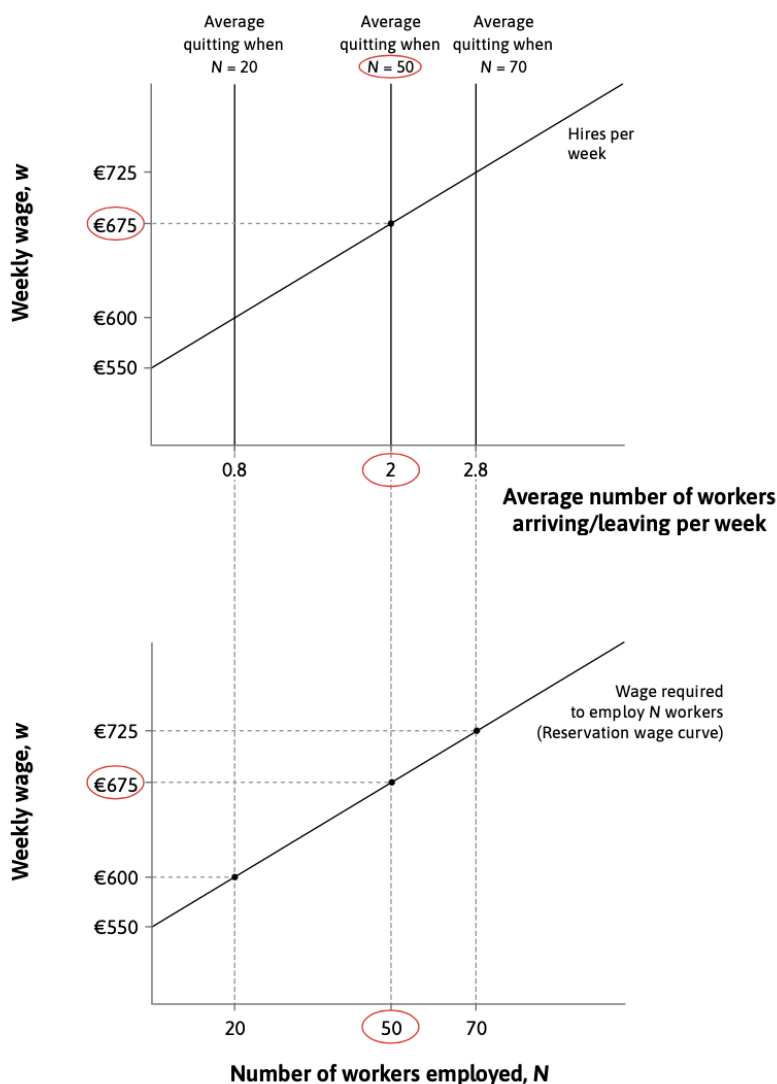
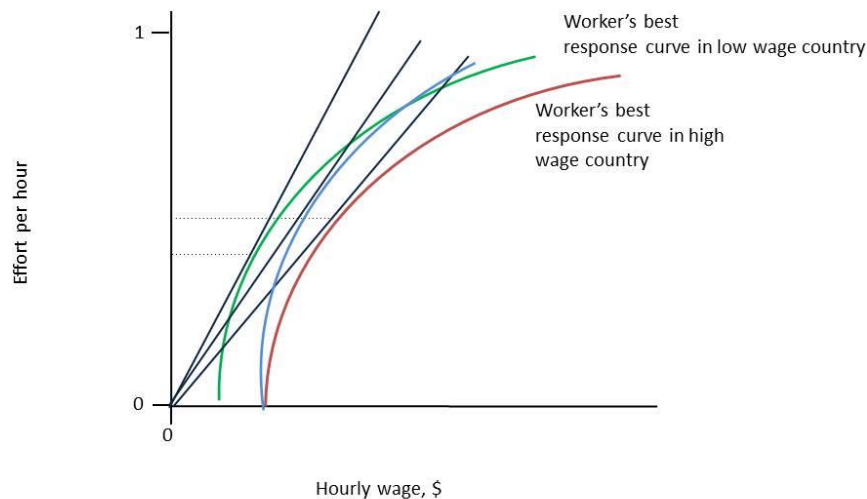


Figure 6.6

4. At the start of this unit, we discussed the decision by many clothing companies to outsource production to Bangladesh and other low-wage economies. Show your results in a single diagram.
 - a. Draw the best response curve of the workers in the high-wage home country in the absence of outsourcing (with the wage on the horizontal axis, and effort on the vertical axis).
 - b. In the same diagram show the best response curve of workers in the foreign low-wage country in the absence of outsourcing. (Assume that wages are measured in dollars in both cases.)
 - c. Show in your diagram what the home country employer will pay home country workers if outsourcing is not possible.
 - d. Show in your diagram what the home country employer will pay workers in the low-wage country if it switches production there (ignore the costs of moving production).
 - e. Now assume that outsourcing is possible and is widely practiced by many firms in the clothing industry. Show the best response function for home country workers under these conditions. Explain why this is different from your answer to 1.

f. Show these outcomes in a diagram.

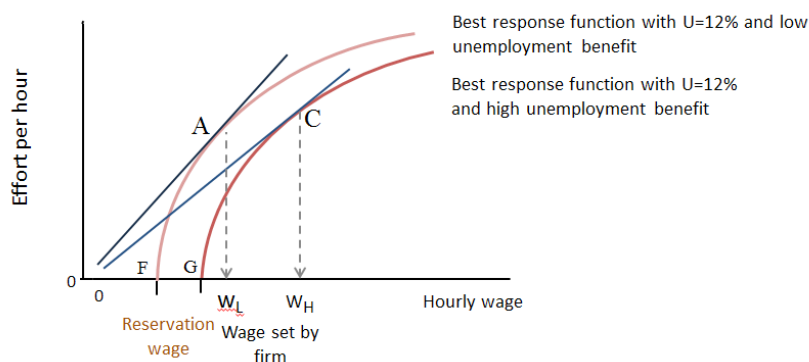
Effects of Outsourcing



2. MULTIPLE-CHOICE QUESTIONS

5. Which of the following statements best describes the game played by the employer and the employee in the labour discipline model?
 - a. The game is a simultaneous game in which the employer chooses the wage level and the employee chooses the effort level simultaneously.
 - b. The game is a one-off game in which the wage and effort levels are determined once and for all.
 - c. The worker selects the effort level that balances his desire to keep his job with his desire to not exhaust himself on the job.
 - d. The employer will attempt to maximise the firm's profits by offering a wage equal to the worker's reservation wage.

6. The figure below depicts the effect of an increase in the unemployment benefit on the workers' best response curve, when the unemployment rate is 12%. Which of the following statements is correct following a rise in the unemployment benefit, *ceteris paribus*?

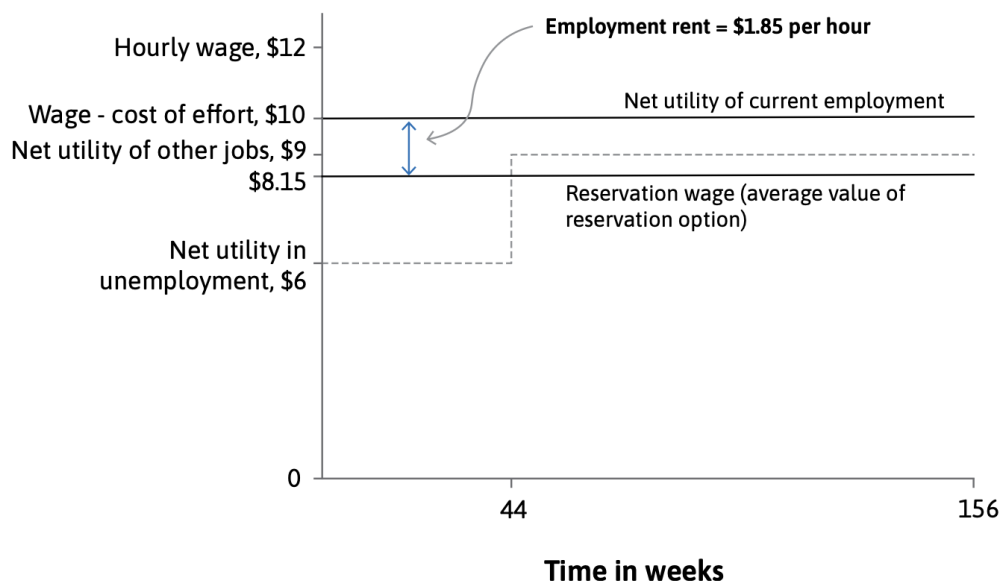


- a. The firm's profit level falls.
 - b. The workers' reservation wage falls.
 - c. The efficiency wage at any given level of employment falls.
 - d. The firm sets a lower efficiency wage.
7. Which of the following statements are correct regarding the difference between markets and firms?
- a. Transactions within firms operate as contracts. Market transactions are not contracts.
 - b. Working in a firm involves accumulation of firm-specific assets that will be lost if the connection to the firm is severed.
 - c. Markets involve a concentration of power, while firms represent a decentralization of economic power.
 - d. It is possible to employ division of labour within firms, while it is not possible in markets.
8. Which of the following statements is correct regarding involuntary unemployment?
- a. When a firm sacks a worker against her wishes, but she can get another job immediately, there is involuntary unemployment.
 - b. When there is involuntary unemployment, the employment rent is zero.
 - c. When there is no involuntary unemployment, firms must be employing workers without ever dismissing them.
 - d. There must be involuntary unemployment in an efficiency wage equilibrium.

3. NUMERICAL PROBLEMS

9. As in all economic models, our simplified representation of Maria's employment rent has deliberately omitted some aspects of the problem that might be important. For example, we have assumed that:
- a. Maria finds a job with a lower pay after her spell of unemployment. (We have calculated Maria's reservation wage using the 'expected' or 'average' length of unemployment. In practice, finding a job is uncertain: it may take less time, or more. Similarly, when she finds one the pay may be above or below average. Since she doesn't know exactly what will happen, she bases the decision on the average values.)

b. Maria continues to receive unemployment benefits as long as she remains unemployed.

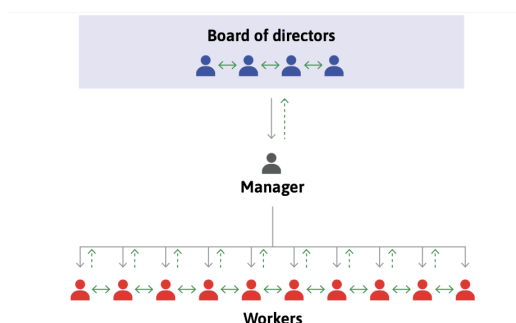


Redraw Figure 6.8b to show how relaxing each of these assumptions would alter the employment rent. Specifically, assume:

- Maria finds a job with the same wage of \$12 per hour after her spell of unemployment.
- Maria's eligibility for unemployment benefits lasts for only 13 weeks.

4. CONCEPTUAL QUESTIONS

10. In Figure 6.1 we showed the actors and decision-making structure of a typical firm. How might the actors and decision-making structure of three organizations, Google, Wikipedia, and a family farm compare with this?



11. Why do you think Mill's vision of a post-capitalist economy of worker-owned cooperatives has not yet occurred?